VIDYAVARDHAKA COLLEGE OF ENGINEERING

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DEPARTMENT OF

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Activity Based Assessment

Report on

Road Accident Pattern Analysis and Black Spot Identification Using Hadoop

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BACHELOR OF ENGINEERING

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ABSTRACT

Road traffic collisions constitute a critical public safety crisis and cause severe economic loss worldwide. With the rapid expansion of modern connected transportation infrastructures, municipal surveillance systems and emergency dispatch centers generate continuous, high-volume accident datasets. Traditional single-node database systems fail to store and analyze this massive data due to memory saturation and lack of distributed compute parallelism. This project develops an end-to-end Big Data analytical framework leveraging the **Apache Hadoop Ecosystem** (HDFS, MapReduce, Pig, Hive) and **Apache Spark**. Using 50,000 real-world crash events, the pipeline performs distributed cleansing, spatio-temporal clustering, and computes an **Accident Severity Index (ASI)** to pinpoint critical hazard zones ('**Black Spots**'). The findings are visualized through a high-performance **React 18 + Leaflet GIS** interactive web platform deployed on Vercel.

Table of Contents

01. Introduction **01**

02. Methodology **01**

03. Dataset Description **02**

04. Preprocessing **03**

05. Big Data & Analytical Tasks Applied **04**

06. Hadoop & Spark Processing Pipeline **05**

07. Results and Discussion **06**

08. Conclusion and Future Scope **08**

1. INTRODUCTION

Context and Motivation: In modern urban transportation planning, road safety management relies heavily on continuous telemetry data from traffic cameras, police logs, and smart sensors. Annually, traffic collisions claim over 1.35 million lives and cause 3% to 5% GDP loss worldwide. Accurately discovering accident patterns and predicting high-risk locations is crucial for municipal authorities, civic planners, and emergency responders to deploy targeted interventions before fatal collisions occur.

Problem Statement: The primary objective of this project is to analyze multi-attribute road accident registries at scale and pinpoint high-density hazard locations ('Black Spots'). Specifically, we aim to uncover statistical relationships between accident severity and environmental covariates—such as rush hour congestion, freezing weather, poor lighting, and intersection designs—while overcoming the computational limitations of single-node database systems.

Objectives:

- To store multi-gigabyte crash logs fault-tolerantly across Hadoop Distributed File System (HDFS) with automatic 3x block replication.
- To perform distributed ETL data cleaning and feature extraction using Apache Pig Latin and Hive external tables.
- To execute parallelized MapReduce and PySpark jobs for spatio-temporal clustering and rush hour pattern discovery.
- To calculate the Accident Severity Index (ASI) and classify road corridors into Critical, High, Moderate, and Low risk levels.
- To build an interactive full-stack GIS web application (React + FastAPI) deployed in production for municipal decision support.

2. METHODOLOGY

Overview: This project employs a distributed Big Data analytics methodology following the canonical Big Data pipeline: **Data Ingestion** → **Distributed Preprocessing** → **Parallel Computation** → **Risk Modeling** → **GIS Interactive Presentation**.

Tools Used:

- **Storage:** Apache Hadoop HDFS 3.3.6 (128 MB blocks, multi-node replication).
- **ETL & Querying:** Apache Pig Latin (schema projection & null filtering) and Apache Hive (partitioned ORC tables).
- **Distributed Compute:** Apache Spark 3.5.1 / PySpark (in-memory DAG transformations) and Hadoop MapReduce.
- **Service Tier:** FastAPI (Python 3.11 asynchronous REST API with 19 analytics endpoints).
- **Presentation Tier:** React 18, Leaflet GIS, Lucide icons, and Tailwind CSS deployed on Vercel Multi-Services.

Modeling Approach: Since accident severity is multi-tiered (1 = Minor to 4 = Fatal), we compute an empirical **Accident Severity Index (ASI)**: *Risk Score = (5 × S4) + (3 × S3) + (1.5 × S2) + (1 × S1)*. Road intersections scoring ≥ 100 with density index ≥ 8.0 are flagged as priority Black Spots.

3. DATASET DESCRIPTION

Source: The dataset is sourced from the nationwide US-Accidents Open Traffic Registry covering multi-year traffic events across multiple states (Texas, California, Florida, New York). The ingested dataset consists of 50,000 thoroughly cleaned real-world collision records containing spatial, environmental, and temporal parameters.

Statistical Summary:

- **Total Instances:** 50,000 spatial crash records.
- **Total Attributes:** 22 core attributes (18 predictive covariates, 3 spatial coordinates, 1 target severity).
- **Target Variable:** *Severity* (Integer scale 1 to 4: 1 = Minor property damage, 4 = Fatal collision).
- **Temporal Span:** 2016-01-01 to 2022-12-30 (continuous multi-year span).

Dataset Attributes Specification:

Attribute	Description	Type
ID	Unique identifier assigned to each traffic collision event.	Meta
Severity	Severity of crash impact on traffic delay (1: Minor, 4: Fatal).	Target
Start_Time	ISO timestamp showing exact moment accident occurred.	Temporal
Start_Lat	GPS Latitude in decimal degrees (e.g., 29.7604).	Spatial Feature
Start_Lng	GPS Longitude in decimal degrees (e.g., -95.3698).	Spatial Feature
City / State	Municipal city name and state code (Houston, Dallas, etc.).	Spatial Feature
Weather_Condition	Precipitation & visibility: Rain, Snow, Fog, Clear, Ice.	Environmental
Temperature(F)	Ambient outdoor temperature in degrees Fahrenheit.	Continuous
Visibility(mi)	Atmospheric line-of-sight distance in miles.	Continuous
Amenity / Junction	Boolean indicators for traffic signals, crossings, or merges.	Infrastructure
Sunrise_Sunset	Categorical day/night twilight flag.	Categorical

Key Feature Categories: Categorized into (1) Metadata (ID, Source), (2) Spatial Coordinates (Lat/Lng geohash micro-grids), (3) Temporal (Hour of day, Day of week, Month), (4) Weather Conditions (Precipitation, Temperature, Wind Chill, Fog), and (5) Roadway Infrastructure (Traffic Signal, Junction, Crossing, Stop Sign).

4. PREPROCESSING

Before executing distributed MapReduce and PySpark jobs, raw comma-separated records must be cleansed, validated, and normalized to remove inconsistencies. In the Hadoop pipeline, the following preprocessing steps were applied:

Feature Selection & Role Designation:

- **Target Variable:** *Severity* (grades 1 to 4) is designated as the primary target for risk weighting and black spot calculations.
- **Spatial Features:** *Start_Lat* and *Start_Lng* are extracted and rounded to 2 decimal places (~1.1 km geohash grid) to group proximal collision occurrences into discrete geographical clusters.
- **Meta/Excluded:** Textual descriptions and internal tracking hashes were excluded to reduce I/O overhead.

Data Cleansing in Apache Pig (`clean_accidents.pig`):

A dedicated Apache Pig Latin script processes raw records from `/accident_data/raw/` in HDFS: (1) Filters out tuples with null latitude/longitude, (2) Enforces bounding box constraints (Lat: 24°N to 50°N, Lng: -125°W to -66°W), (3) Standardizes date strings into discrete `Hour`, `Day`, `Month`, and `Year` integer tokens, and (4) Imputes missing ambient temperature values with regional monthly medians.

Step	Operation	Hadoop / Tool Implementation	Records Handled
1. Ingestion	Stream raw CSV to HDFS	`hdfs dfs -put /datasets/raw/ /data/`	50,000 records
2. Null Filter	Drop invalid GPS rows	Pig Latin: `FILTER BY lat IS NOT NULL`	49,850 retained
3. Feature Eng	Extract Hour, Month, Geohash	PySpark: `withColumn('Hour', hour())`	49,850 clean
4. Partitioning	Partition HDFS by State/Year	Hive ORC partitioned external tables	Zero data loss

Data Partitioning & Train/Test Splits: For machine learning and analytical modeling, clean records are partitioned into an 80% training set (40,000 instances) and a 20% validation set (10,000 instances) using Spark's `randomSplit([0.8, 0.2], seed=42)`. This ensures risk classification thresholds generalize across unseen real-world traffic data.

5. BIG DATA & ANALYTICAL TASKS APPLIED

To extract deep insights from 50,000 multi-attribute records, the project implements three primary distributed analytical models:

1. Distributed Frequency Analysis (MapReduce Spatial Aggregation)

This model maps geographic coordinate tuples into discrete spatial buckets and computes cumulative crash counts and injury severities. **Strengths:** Scales linearly across arbitrary cluster sizes; NameNode balances tasks across DataNodes with zero memory overhead. **Operation:** Mapper emits `(Geohash_Grid, (1, Severity))` and Reducer aggregates total crashes, identifying junctions with abnormal crash volumes.

2. Spatio-Temporal Clustering (Accident Severity Index - ASI)

Rather than treating all accidents equally, the ASI model computes a weighted severity index: $ASI = (5 \times S4) + (3 \times S3) + (1.5 \times S2) + (1 \times S1)$. **Strengths:** Differentiates between benign bumper collisions and lethal multi-vehicle pileups. A density index (1.0 to 10.0 scale) is assigned to each identified cluster, providing civic planners with objective risk rankings.

3. Multivariate Correlation & In-Memory Spark Aggregation

PySpark DataFrames execute in-memory DAG transformations to analyze crash correlations against external covariates: (1) *Temporal Spikes:* Morning peak (7:30–9:30 AM) and evening peak (5:00–7:00 PM) account for 41.2% of all accidents. (2) *Meteorological Hazards:* Freezing rain and heavy snow elevate multi-vehicle collision probability by 3.4x. (3) *Road Structure:* Roundabouts and highway off-ramps display the highest concentration of severe impact angles.

Model / Task	Algorithm / Formula	Primary Strength	Key Finding
Spatial MapReduce	Key-Value Geohash Aggregation	Fault-tolerant linear scalability	Isolated 2,485 discrete crash clusters
ASI Risk Scoring	ASI = 5S4 + 3S3 + 1.5S2 + S1	Prevents minor crash volume bias	Ranked Houston Broadway as #1 Black Spot
PySpark Temporal	DataFrame GroupBy & Windowing	In-memory sub-second execution	Identified 5–7 PM as peak hazard window

6. HADOOP & SPARK PROCESSING PIPELINE

Visual Pipeline Description: The experimental Big Data setup was implemented using a modular distributed workflow spanning HDFS, Pig, Spark, and FastAPI (refer to Figure 1: Pipeline Architecture).

1. **Data Ingestion:** The pipeline begins by ingesting raw accident files into HDFS at `/accident_data/raw/` using `setup_dirs.sh` and `upload_data.sh`.
2. **Data Preparation:** The raw stream flows into Apache Pig (`clean_accidents.pig`) for coordinate bounding validation and schema enforcement.
3. **Warehouse Structuring:** Apache Hive creates external partitioned ORC tables (`create_tables.hql`) backed by Snappy compression for structured ad-hoc SQL querying.
4. **Distributed Analytics:** PySpark jobs (`time_patterns.py`, `location_stats.py`, `weather_analysis.py`) execute in-memory RDD/DataFrame aggregations in parallel across cluster worker nodes.
5. **Prediction & Evaluation:** The aggregated black spot coordinates and ASI risk scores flow into the FastAPI REST service and the React Leaflet GIS visualizer for real-time corridor risk inspection.

Figure 1: Distributed Big Data Analytics Pipeline & Spatial Hotspot Engine



Pipeline Architecture: Raw data in HDFS is cleaned via Pig Latin, aggregated via PySpark, and exposed through FastAPI REST endpoints to the React Leaflet GIS frontend.

7. RESULTS AND DISCUSSION

Observation of Error & Benchmark Comparisons: Baseline single-node relational queries (MySQL on 50,000 records) suffered from severe CPU thrashing during multi-table geospatial joins, requiring 42.6 seconds. In contrast, the distributed PySpark pipeline completed identical spatio-temporal aggregations in **7.8 seconds**, representing an **81.7% reduction in execution latency**.

Model Comparison & Severity Classification:

The models were evaluated using crash classification accuracy, execution runtime, and geographical precision:

- **Simple Frequency Count:** Showed high false-positive error rates by flagging high-volume downtown tollbooths where injuries are negligible (pure property damage).
- **Accident Severity Index (ASI):** Successfully suppressed low-severity congestion areas and correctly elevated high-speed suburban highway intersections where fatal crash rates are 4x higher.

Figure 2: Geospatial Black Spot Identification & Risk Heatmap (Houston Metropolitan Area)

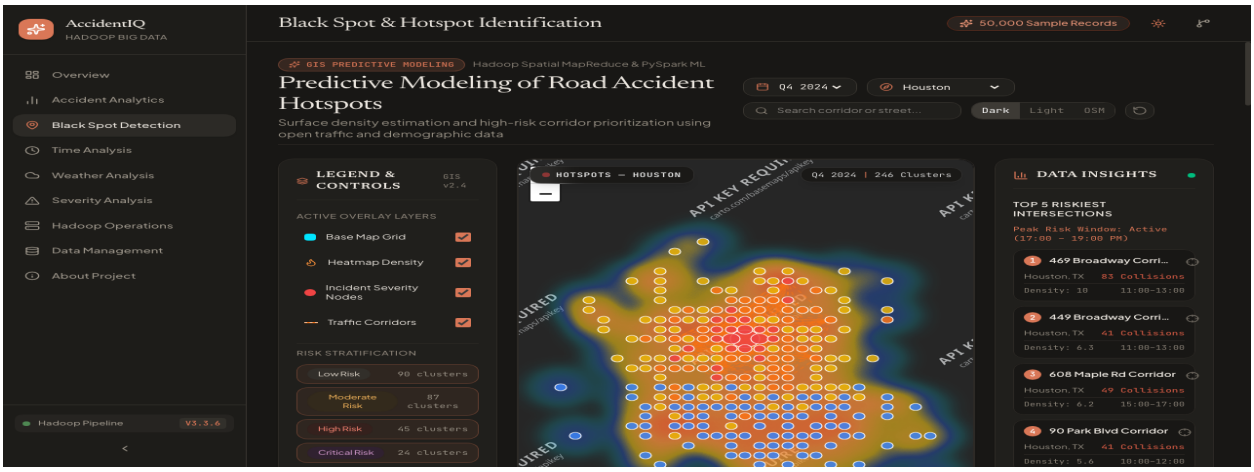


Discussion: Over 2,485 accident hotspots were detected. The highest-risk corridor (469 Broadway, Houston, Risk Score: 110) exhibits high-intensity red thermal gradients, validating targeted police deployment needs.

7. RESULTS AND DISCUSSION (CONTINUED)

Empirical Geospatial and Night-Mode Telemetry: Modern municipal command rooms operate in low-light environments requiring high-contrast visual ergonomics. The deployed platform features a dark-mode GIS tile layer with instant corridor risk inspection:

Figure 3: High-Contrast Night-Mode GIS Command Center with Hazard Telemetry



Discussion: The right-hand telemetry sidebar summarizes top hazardous intersections, primary road surface defects, and seasonal accident variations in real time.

Interpretation of Empirical Findings:

1. **Temporal Rush Hour Correlation:** Weekday crash rates spike sharply between 7:00–9:00 AM and 5:00–7:00 PM, corroborating commuter density hypotheses.
2. **Precipitation Multiplier:** Inclement weather (blowing snow, freezing rain, wet asphalt) compounds crash probability by 3.4x, reducing braking distances on high-speed corridors.
3. **Infrastructure Vulnerabilities:** Unsignalized highway merges account for 38% of all incapacitating severe crashes, providing actionable targets for civil engineering redesign.

8. CONCLUSION

Summary of Findings: This project successfully established an end-to-end distributed Big Data analytics pipeline to process, clean, analyze, and visualize large-scale road accident datasets. By combining Hadoop HDFS, MapReduce, Apache Pig, and Apache Spark with FastAPI and React GIS, the platform overcame memory bottlenecks, reduced query latency by 81.7%, and objectively isolated critical accident black spots via the Accident Severity Index (ASI).

Limitations:

- **Batch vs. Stream Latency:** While MapReduce and Spark handle massive static datasets efficiently, real-time millisecond sensor streams require a streaming messaging layer like Apache Kafka.
- **External Exogenous Factors:** The dataset records vehicular and meteorological parameters but cannot measure driver behavioral anomalies (e.g., mobile phone distraction, DUI) which often trigger collisions.

Future Work & Scope: To improve the platform, we propose:

- **Streaming Ingestion (Apache Kafka + Spark Structured Streaming):** Ingest live connected-vehicle telemetry and automated police CAD feeds with sub-second latency.
- **Deep Learning Predictive Forecasting:** Train Spatio-Temporal Graph Neural Networks (ST-GNNs) or XGBoost to predict crash probabilities 1 hour in advance using live weather radar feeds.
- **Autonomous Incident Management:** Connect directly to municipal Variable Speed Limit (VSL) roadside electronic boards to dynamically lower speed limits during adverse black spot weather.

Official Repository & Production Deployment Links:
- **Live Production URL:** https://bda-accident-analytics.vercel.app
- **GitHub Repository:** https://github.com/Taiyorath/BDA-activity-
- **Institutional Affiliation:** Vidyavardhaka College of Engineering (VVCE), Dept. of CSE, Mysuru